



**Working Draft
MEF W160 v0.4**

LSO Cantata and LSO Sonata Product Offering Availability and Pricing Discovery API - Developer Guide

This draft represents MEF work in progress and is subject to change.

January 2025

EXPORT CONTROL: This document contains technical data. The download, export, re-export or disclosure of the technical data contained in this document may be restricted by applicable U.S. or foreign export laws, regulations and rules and/or applicable U.S. or foreign sanctions ("Export Control Laws or Sanctions"). You agree that you are solely responsible for determining whether any Export Control Laws or Sanctions may apply to your download, export, reexport or disclosure of this document, and for obtaining (if available) any required U.S. or foreign export or reexport licenses and/or other required authorizations.

Disclaimer

© MEF Forum 2025. All Rights Reserved.

The information in this publication is freely available for reproduction and use by any recipient and is believed to be accurate as of its publication date. Such information is subject to change without notice and MEF Forum (MEF) is not responsible for any errors. MEF does not assume responsibility to update or correct any information in this publication. No representation or warranty, expressed or implied, is made by MEF concerning the completeness, accuracy, or applicability of any information contained herein and no liability of any kind shall be assumed by MEF as a result of reliance upon such information.

The information contained herein is intended to be used without modification by the recipient or user of this document. MEF is not responsible or liable for any modifications to this document made by any other party.

The receipt or any use of this document or its contents does not in any way create, by implication or otherwise:

- (a) any express or implied license or right to or under any patent, copyright, trademark or trade secret rights held or claimed by any MEF member which are or may be associated with the ideas, techniques, concepts or expressions contained herein; nor
- (b) any warranty or representation that any MEF member will announce any product(s) and/or service(s) related thereto, or if such announcements are made, that such announced product(s) and/or service(s) embody any or all of the ideas, technologies, or concepts contained herein; nor
- (c) any form of relationship between any MEF member and the recipient or user of this document.

Implementation or use of specific MEF standards, specifications or recommendations will be voluntary, and no Member shall be obliged to implement them by virtue of participation in MEF Forum. MEF is a non-profit international organization to enable the development and worldwide adoption of agile, assured and orchestrated network services. MEF does not, expressly or otherwise, endorse or promote any specific products or services.

Copyright

© MEF Forum 2025. Any reproduction of this document, or any portion thereof, shall contain the following statement: "Reproduced with permission of MEF Forum." No user of this document is authorized to modify any of the information contained herein.

Table of Contents

- List of Contributing Members
- 1. Abstract
- 2. Terminology and Abbreviations
- 3. Compliance Levels
- 4. Introduction
 - 4.1. Description
 - 4.2. Conventions in the Document
 - 4.3. Relation to Other Documents
 - 4.4. Approach
 - 4.5. High-Level Flow
- 5. API Description
 - 5.1. High-level use cases
 - 5.2. API Endpoint and Operation Description
 - 5.3. Specifying the Buyer ID and the Seller ID
 - 5.4. Integration of Product Specifications into the API
 - 5.5. Sample Product Specification
 - 5.6. Model Structural Validation
 - 5.7. Security Considerations
- 6. API Interaction & Flows
 - 6.1. Use case 1: Retrieve Product Offering Availability
 - 6.1.1. Request
 - 6.1.2. Response
 - 6.2. Use case 2: Retrieve Pricing for a Product Offering Configuration
 - 6.2.1. Request
 - 6.2.2. Response
- 7. API Details
 - 7.1. API patterns
 - 7.1.1. Indicating errors
 - 7.1.1.1. Type Error
 - 7.1.1.2. Type Error400
 - 7.1.1.3. `enum` Error400Code
 - 7.1.1.4. Type Error401
 - 7.1.1.5. `enum` Error401Code
 - 7.1.1.6. Type Error403
 - 7.1.1.7. `enum` Error403Code
 - 7.1.1.8. Type Error422
 - 7.1.1.9. `enum` Error422Code
 - 7.1.1.10. Type Error500
 - 7.2. API Data model
 - 7.2.1. Product Offering Availability
 - 7.2.1.1. Type ProductOfferingAvailability_Request
 - 7.2.1.2. Type ProductOfferingAvailability
 - 7.2.1.3. Type ProductOfferingConfiguration
 - 7.2.1.4. Type ProductOfferingRef
 - 7.2.1.5. Type MEFProductConfiguration
 - 7.2.1.6. Type ProductSpecificationRef
 - 7.2.2. Pricing Discovery
 - 7.2.2.1. Type PricingDiscovery_Request
 - 7.2.2.2. Type PricingDiscovery
 - 7.2.2.3. Type PricingAndTerm
 - 7.2.2.4. Type QuotePrice
 - 7.2.2.6. Type MEFItemTerm

- 7.2.2.7. **enum** MEFEndOfTermAction
- 7.2.2.8. **enum** MEFPriceType
- 7.2.2.9. Type Money
- 7.2.2.10. Type Price
- 7.2.3. Place representation
 - 7.2.3.1. Type RelatedPlaceRefWithSubUnit
 - 7.2.3.2. Type PlaceRef
 - 7.2.3.3. Type GeographicAddressRef
 - 7.2.3.4. Type GeographicSiteRef
 - 7.2.3.5. Type SubUnit
- 7.2.4. Common Types
 - 7.2.4.1. Type Duration
 - 7.2.4.2. **enum** ProductActionAddModifyType
 - 7.2.4.3. Type ProductRef
 - 7.2.4.4. Type ProductRelationship
 - 7.2.4.5. **enum** TimeUnit
- 8. References

List of Contributing Members

The following members of the MEF participated in the development of this document and have requested to be included in this list.

Member

Table 1. Contributing Members

1. Abstract

This standard is intended to assist implementation of the Product Offering Availability and Pricing Discovery functionality defined for the LSO Cantata and LSO Sonata Interface Reference Points (IRPs), for which requirements and use cases are defined in MEF 110 *Product Offering Availability and Pricing Discovery - Business Requirements and Use Cases* [MEF 110]. This standard consists of this document and complementary API definitions.

This standard normatively incorporates the following files by reference as if they were part of this document, from the GitHub repository:

<https://github.com/MEF-GIT/MEF-LSO-Sonata-SDK>

- `productApi/availabilityAndPricingDiscovery/productOfferingAvailabilityAndPricingDiscovery.api.yaml`

<https://github.com/MEF-GIT/MEF-LSO-Cantata-SDK>

- `productApi/availabilityAndPricingDiscovery/productOfferingAvailabilityAndPricingDiscovery.api.yaml`

2. Terminology and Abbreviations

This document does not define any new terms or definitions. All of them are defined in the standards referenced below and are included in this document by reference:

- [\[MEF 110\]](#)
- [\[MEF 55.1\]](#)
- [\[MEF 55.1.1\]](#)
- [\[MEF 80\]](#)
- [\[MEF 150\]](#)

3. Compliance Levels

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 ([RFC2119], [RFC8174]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as **[Ox]** for optional.

A paragraph preceded by **[CRa]<** specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, "**[CR1]<[D38]**" indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by **[CDb]<** specifies a Conditional Desirable Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by ****[COc]<**** specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

4. Introduction

This standard specification document describes the Application Programming Interface (API) for Product Offering Availability and Pricing Discovery functionality of the LSO Cantata Interface Reference Point (IRP) and Sonata IRP as defined in the MEF 55.1 *Lifecycle Service Orchestration (LSO): Reference Architecture and Framework* [MEF55.1]. The LSO Reference Architecture is shown in Figure 1 with both IRPs highlighted.

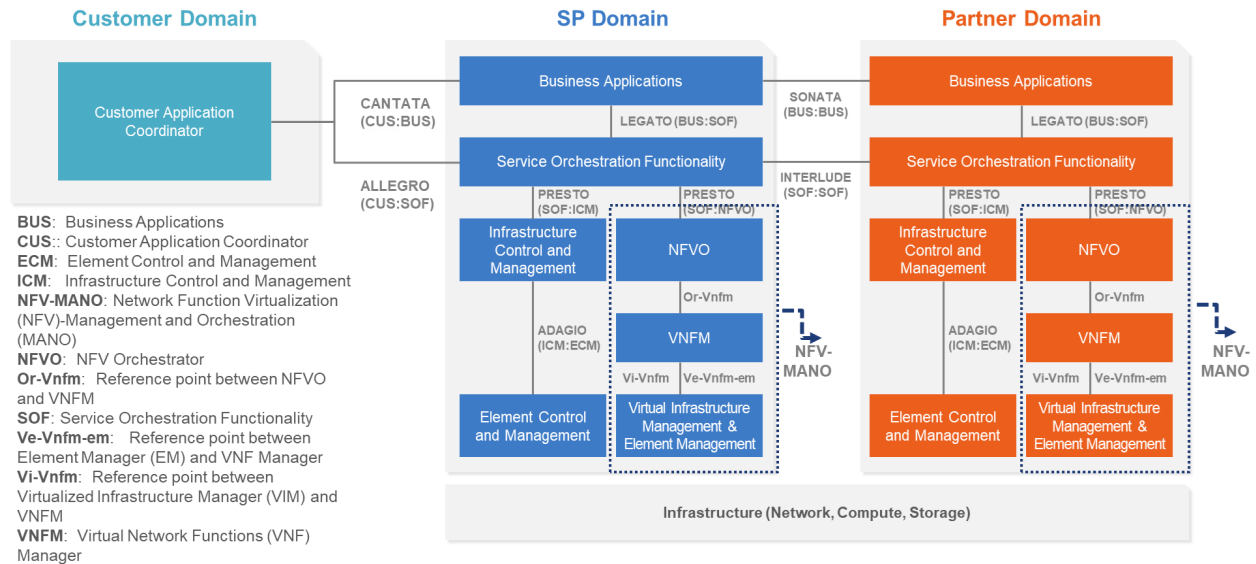


Figure 1. The LSO Reference Architecture

This document is structured as follows:

- **Chapter 4** provides an introduction to Product Offering Availability and Pricing Discovery and its description in a broader context of Cantata and Sonata
- **Chapter 5** gives an overview of endpoints, resource model and design patterns.
- Use cases and flows are presented in **Chapter 6**.
- And finally, **Chapter 7** complements previous sections with a detailed resource model description.

4.1. Description

As specified in [MEF110]:

Product Offering Availability Discovery is a way for the Buyer to specify a Product Specification and Delivery Context to the Seller in order to receive from the Seller a list of zero or more Product Offering Configurations.

Pricing Discovery allows the Buyer to receive from the Seller a list of one or more Pricing and term information for the Product Configuration and Delivery Context specified by the Buyer.

Product Offering Availability and Pricing Discovery are intended for use by a Buyer who has already determined that they will use a particular Seller for products at a particular location/UNI and are only wanting to know exactly what products and configurations are available there, and what pricing structures are available for those products. Product Offering Availability and Pricing Discovery are not intended to be used to determine whether the Seller can provide products meeting the Buyers needs at that location/UNI in the first place (i.e. the assumption is that they can) - since it is not possible to retrieve complete information about, for example, the total cost of

a set of related products before ordering some of them. The Quote mechanism defined in [MEF80] is more appropriate for that type of use.

As with any other interaction between a Buyer and Seller, several prerequisites must be fulfilled. This is done during onboarding and includes, but is not limited to the following:

- Any elements of the Buyer that are opaque to End Customers, such as ENNIs are in place and are available for use.
- The period of time after which auto-renewal occurs and in which the Buyer can disconnect the Product without penalty is agreed to by the Buyer and Seller.
- The pricing framework has been agreed to by the Buyer and Seller.
- If `businessHours` and `businessDays` are used as values for the `TimeUnit` attribute, the Buyer and Seller must agree to their definition
- The Seller's right to reuse a `productOfferingConfigurationIdentifier` for different Delivery Contexts is agreed to by the Buyer and Seller.
- The `productConfiguration` attributes that are returned in the Seller's response to a Product Offering Availability Discovery request are agreed.
- Any referenced pre-requisite Products are in place and are in the correct Administrative or Operational state to be referenced.

Details of how onboarding happens, and the agreements and data exchange that happens through the onboarding process, are outside the scope of this document.

The `productConfiguration` attributes that are returned by the Seller could be different when the Action is `add` versus `modify`.

Delivery Context is a set of related Products and Places that are associated with a Product. The possible and/or required relations are defined in each of the Product Specification Standards. For example, the Delivery Context of a UNI would be the physical location at which it is installed, whereas the Delivery Context for an Access E-Line would be the UNI and ENNI that it connects.

4.2. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. `ProductOfferingAvailability`).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.
- In the API details tables and UML diagrams required attributes are marked with a `*` next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

4.3. Relation to Other Documents

The requirements and use cases for Product Offering Availability and Pricing Discovery functionality are defined in MEF 110 [MEF110].

4.4. Approach

As presented in Figure 2. both LSO Cantata and LSO Sonata API frameworks consist of three structural components:

- Generic API framework

- Product-independent information (Function-specific information and Function-specific operations)
- Product-specific information (MEF product specification data model)

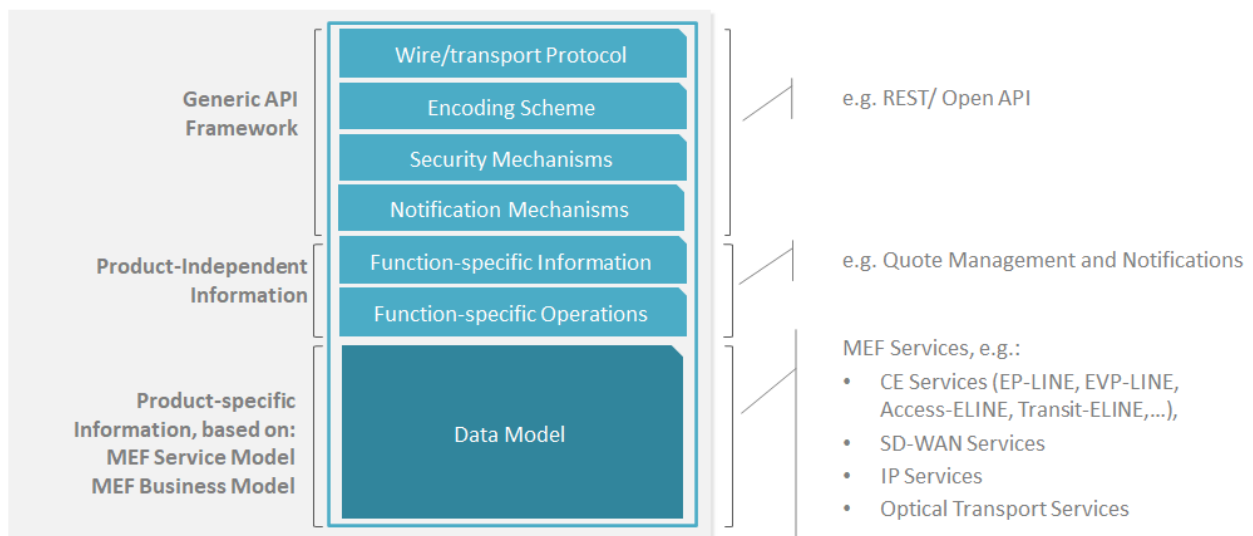


Figure 2. Cantata and Sonata API framework

The essential concept behind the framework is to decouple the common structure, information, and operations from the specific product information content.

Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all Cantata or Sonata APIs.

Secondly, the product-independent information of the framework focuses on a model of a particular Cantata or Sonata functionality and is agnostic to any of the product specifications.

Finally, the product-specific information part of the framework focuses on MEF product specifications that define business-relevant attributes and requirements for trading MEF subscriber and MEF operator services.

4.5. High-Level Flow

Product Offering Availability and Pricing Discovery is part of a broader Cantata and Sonata End-to-End flow. Figure 3. below shows a high-level diagram to get a good understanding of the whole process.



Figure 3. Cantata and Sonata End-to-End Function Flow

- Address Validation:
 - Allows the Buyer to retrieve address information from the Seller, including exact formats, for addresses known to the Seller.
- Site Retrieval:

- Allows the Buyer to retrieve Service Site information including exact formats for Service Sites known to the Seller.
- Product Catalog:
 - Allows the Buyer to retrieve Product Categories and Product Offering information, including specification details from a Seller's Product Catalog.
- Product Offering Qualification (POQ):
 - Allows the Buyer to check whether the Seller can deliver a product or set of products from among their product offerings at the geographic address or a service site specified by the Buyer; or modify a previously purchased product.
- Quote:
 - Allows the Buyer to submit a request to find out how much the installation of an instance of a Product Offering, an update to an existing Product, or a disconnect of an existing Product will cost.
- Product Order:
 - Allows the Buyer to request the Seller to initiate and complete the fulfillment process of an installation of a Product Offering, an update to an existing Product, or a disconnect of an existing Product at the address defined by the Buyer.
- Product Inventory:
 - Allows the Buyer to retrieve information about existing Product instances from Seller's Product Inventory.
- Billing:
 - Allows the Seller to generate the document to the Buyer relating to charges associated with Products provided by the Seller to the Buyer.
- Trouble Ticketing:
 - Allows the Buyer to create, retrieve, and update Trouble Tickets as well as receive notifications about Incidents' and Trouble Tickets' updates. This allows for managing issues and situations that are not part of the normal operations of the Product provided by the Seller.

Product Offering Availability and Pricing Discovery APIs are drawn as a fork to standard POQ and Quote as they implement the same functionality but in a slightly different approach. The discrepancies will be explained in detail in further sections.

5. API Description

This section discusses the API structure and design patterns. It starts with the high-level use cases diagram and then it describes the REST endpoints with use case mapping.

5.1. High-level use cases

Figure 4 presents a high-level use case diagram as specified in MEF 110 [MEF110]. This picture aims to help understand endpoint mapping. Use cases are described extensively in [chapter 6](#)

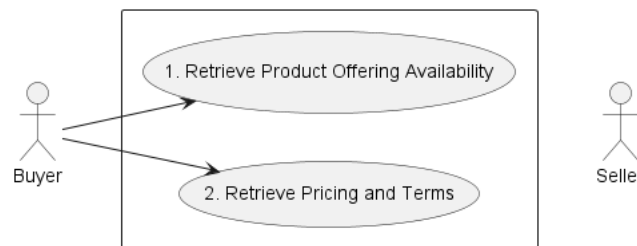


Figure 4. High-level use cases

5.2. API Endpoint and Operation Description

Base URL for Cantata:

```
https://{serverBase}:{port}
{{?/seller_prefix}}/mefApi/cantata/productOfferingAvailabilityAndPricingDiscovery/v4/
```

Base URL for Sonata:

```
https://{serverBase}:{port}
{{?/seller_prefix}}/mefApi/sonata/productOfferingAvailabilityAndPricingDiscovery/v4/
```

Note: All examples will include only the Sonata version of the Base Path.

Table 2 lists the API endpoints with mapping to business use cases:

API endpoint	Description	MEF 110 Use case mapping
POST /productOfferingAvailability	A request initiated by the Buyer to receive a list of zero or more Product Offering Configurations	UC 1: Retrieve Product Offering Availability
POST /pricingDiscovery	A request initiated by the Buyer to receive a list of one or more pricing and terms information for the Product Offering Configuration	UC 2: Retrieve Pricing and Terms

Table 2. Seller side endpoints.

[R1] The Buyer implementation **MUST** be able to use all REST methods that are listed in Table 2. [MEF110 R1], [MEF110 R2], [MEF110 R24], [MEF110 R25]

5.3. Specifying the Buyer ID and the Seller ID

A business Entity willing to represent multiple Buyers or multiple Sellers must follow requirements of [MEF 150] chapter 8.8, which states:

For requests of all types, there is a business Entity that initiates an Operation (called a Requesting Entity) and a business Entity that is responding to this request (called the Responding Entity). In the simplest case, the Requesting Entity is the Buyer and the Responding Entity is the Seller. However, in some cases, the Requesting Entity may represent more than one Buyer and similarly, the Responding Entity may represent more than one Seller.

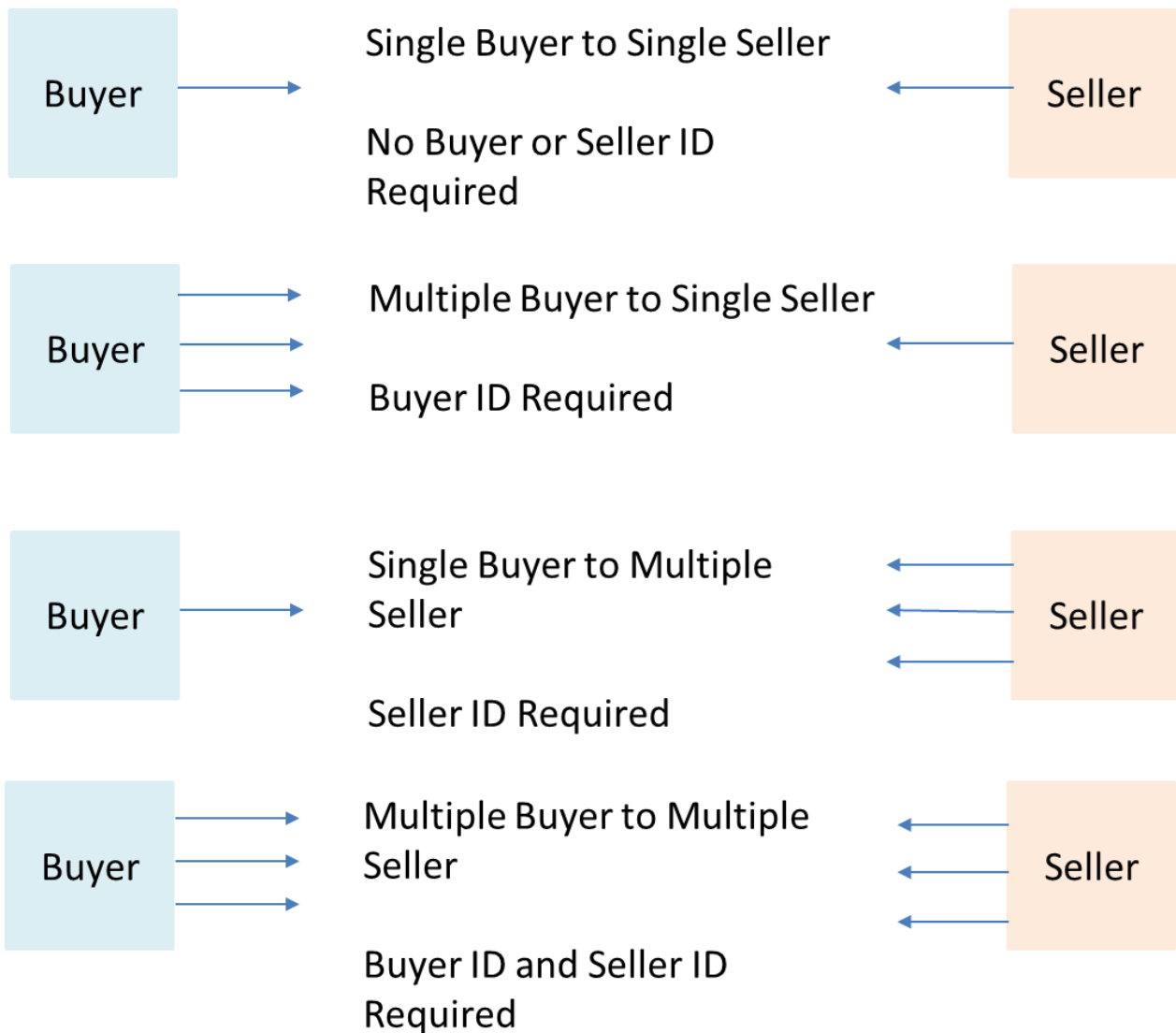


Figure 5. Buyer ID and Seller ID Examples

As shown in Figure 5, if a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing a single Seller, Buyer and Seller IDs are not required to be passed between the two Entities. If a Requesting Entity representing more than one Buyer is doing business with a Responding Entity representing a single Seller, the Buyer ID is required to be passed between the two Entities. If a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing multiple Sellers, the Seller ID is required to be passed between the two Entities. If a Requesting Entity representing multiple Buyers is doing business with a Responding Entity representing multiple Sellers, both the Buyer ID and the Seller ID are required to be passed between the Entities.

While it is outside the scope of this specification, it is assumed that the Requesting Entity and the Responding Entity are aware of each other and can authenticate requests initiated by the other party. It is further assumed that the Buying Entity knows:

- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity; and
- the list of Sellers that this Responding Entity represents to this Requesting Entity.

It is also assumed that the Responding Entity knows:

- the list of Sellers that this Responding Entity represents to this Requesting Entity and
- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity.

In the API the **buyerId** and **sellerId** are represented as optional query parameters in each operation defined.

[R2] If the Requesting Entity has the authority to represent more than one Buyer the request **MUST** include **buyerId** that identifies the Buyer being represented [MEF150 R62]

[R3] If the Responding Entity represents more than one Seller to this Buyer the request **MUST** include **sellerId** that identifies the Seller with whom this request is associated [MEF150 R63]

5.4. Integration of Product Specifications into the API

Product specification schemas are defined and provided using JsonSchema (draft 7) format and are integrated into the API using the extension pattern.

The extension hosting type in the API data model is **MEFProductConfiguration**. The **@type** attribute of that type must be set to a value that uniquely identifies the product specification. A unique identifier for MEF standard product specifications is in URN format and is assigned by MEF. This identifier is provided as root schema **\$id** and in product specification documentation. Use of non-MEF standard product definitions is allowed. In such a case the schema identifier must be agreed upon between the Buyer and the Seller.

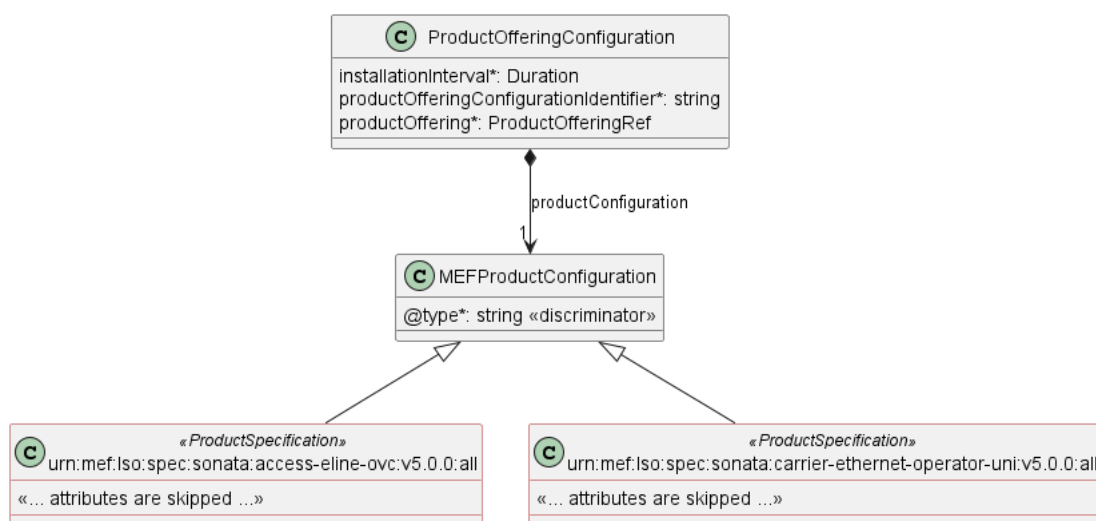


Figure 6. The Extension Pattern

Figure 6 presents two MEF **<<ProductSpecifications>>** that represent Access E-Line OVC and Carrier Ethernet Operator UNI products. When these products are used as a Product Offering Availability or Pricing Discovery payload the **@type** of **MEFProductConfiguration** takes **"urn:mef:iso:spec:sonata:access-eline-ovc:v5.0.0:all"** or

"urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all" value to indicate which product specification schema must be used to interpret a set of product-specific attributes included in the payload. An example of a product definition inside the `productConfiguration` is presented in [Section 6.1.2](#).

The *all* suffix after the product type name in the URN indicates that the schema can be used in the context of all APIs (POQ, Quote, Order, or Inventory).

The example below shows a header of a Product Specification schema, which is referring to the Access E-Line OVC, where "\$id": urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all is the abovementioned URN:

```
'$schema': http://json-schema.org/draft-07/schema#
'$id': urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all
title: MEF LSO Sonata - Access Eline OVC Product Schema
```

Product specifications are provided as Json schemas without the `MEFProductConfiguration` context.

Product-specific attributes are introduced via the `ProductOfferingConfiguration`. This entity has the `productConfiguration` attribute of type `MEFProductConfiguration` which is used as an extension point for product-specific attributes.

Implementations might choose to integrate selected product specifications into the data model during development. In such a case an integrated data model is built and product specifications are in an inheritance relationship with `MEFProductConfiguration` as described in the OAS specification. This pattern is called **Static Binding**. The SDK is additionally shipped with a set of API definitions that statically bind all product-related APIs (POQ, Quote, Order, Inventory) with all corresponding product specifications available in the release. The snippet below presents an example of a static binding of the envelope API with two product specifications. This is a subset of one yaml file describing the API. Some attributes and their descriptions are truncated for readability.

```
MEFProductConfiguration:
  description:
    MEFProductConfiguration is used as an extension point for MEF-specific
    product/service payload. The '@type' attribute is used as a discriminator
  discriminator:
  mapping:
    urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all: '#/components/schemas/AccessElineOvc'
    urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all:
    '#/components/schemas/CarrierEthernetOperatorUni'
  propertyName: '@type'
  properties:
    '@type':
      description:
        The name of the type, defined in the JSON schema specified above, for
        the product that is the subject of the Request. The named type must be
        a subclass of MEFProductConfiguration.
      type: string
AccessElineOvc:
  allOf:
    - $ref: '#/components/schemas/MEFProductConfiguration'
    - $ref: '#/components/schemas/AccessElineOvcCommon'
    - type: object
  properties:
    uniEp:
      $ref: '#/components/schemas/AccessElineOvcEndPoint'
      description:
        MEF 26.2 sec. 16 - The OVC EP object for the OVC EP at the UNI. The
        UNI OVC End Point must be included in the Access E-Line Product.
    enniEp:
      $ref: '#/components/schemas/AccessElineOvcEndPoint'
      description:
```



```

MEF 26.2 sec. 16 - The OVC EP object for the OVC EP at the ENNI.
The ENNI OVC End Point must be included in the Access E-Line
Product.
CarrierEthernetOperatorUni:
  allOf:
    - $ref: '#/components/schemas/MEFProductConfiguration'
    - properties:
        listOfPhysicalLinks:
          type: array
          items:
            $ref: '#/components/schemas/UniPhysicalLink'
          minItems: 1
          uniqueItems: true
        linkAggregation:
          $ref: '#/components/schemas/LinkAggType'
        aggregationLinkMap:
          type: array
          items:
            $ref: '#/components/schemas/ConversationIdToAggregationLinkMap'
          minItems: 0
          uniqueItems: true
        maximumServiceFrameSize:
          type: integer
          minimum: 1522
        ...

```

Alternatively, implementations might choose not to build an integrated model and choose a different mechanism allowing runtime validation of product-specific fragments of the payload. The system can validate a given product against a new schema without redeployment. This pattern is called **Dynamic Binding**.

Regardless of the chosen implementation pattern, the HTTP payload is exactly the same. Both implementation approaches must conform to the requirements specified below.

[R4] **MEFProductConfiguration** type is an extension point that **MUST** be used to integrate product specifications' properties into a request/response payload.

[R5] The **@type** property of **MEFProductConfiguration** **MUST** be used to specify the type of the extending entity.

[R6] Product attributes specified in the payload must conform to the product specification specified in the **@type** property.

5.5. Sample Product Specification

The Sonata SDK contains product specification definitions, from which Access E-Line [MEF 106] is used in the payload samples in this section. They are located in the SDK at:

```

\productSchema\carrierEthernet\operatorEthernet\accessEline\accessElineOvc.yaml
\productSchema\carrierEthernet\operatorEthernet\carrierEthernetOperatorUni\carrierEthernetOperatorUni.yaml

```

The product specification data model definitions are available as JsonSchema (version **draft 7**) documents. Figures 7 and 8 present an instance diagram of OVC and UNI products with simplified configuration. This document aims to explain the pattern of exchanging product-specific attributes, not to explain the particular product itself.

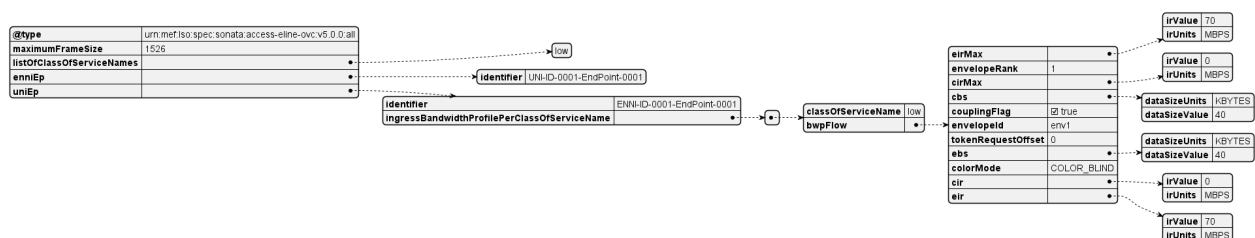


Figure 7. Access E-Line OVC Product Shortened Configuration Example

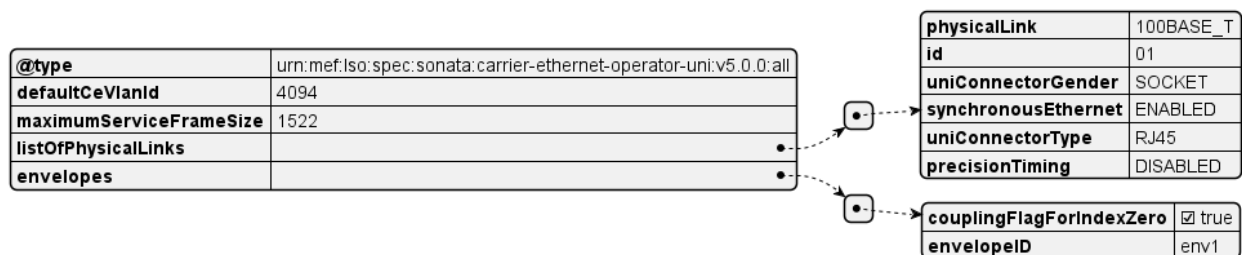


Figure 8. Operator Carrier Ethernet UNI Product Shortened Configuration Example

Product specifications define several product-related and envelope-related requirements. For example:

- for an Access E-Line OVS product two mandatory relationship roles must be specified, one with the operator ENNI (**CONNECTS_TO_ENNI**) and a second with the operator UNI (**CONNECTS_TO_UNI**).
- in the case of a **modify** action, product relationships must have the same value as in the **add** action. They must not be changed
- for an operator UNI product a place relationship (**INSTALL_LOCATION**) must be specified
- in the case of a **modify** action, place relationships must have the same value as in the **add** action. They must not be changed

Figure 9 presents the Access E-Line product required relations. The Access E-Line OVC has two product relationships:

- towards ENNI - **CONNECTS_TO_ENNI**
- towards UNI - **CONNECTS_TO_UNI**

The UNI product has one place relationship pointing to **INSTALL_LOCATION**.

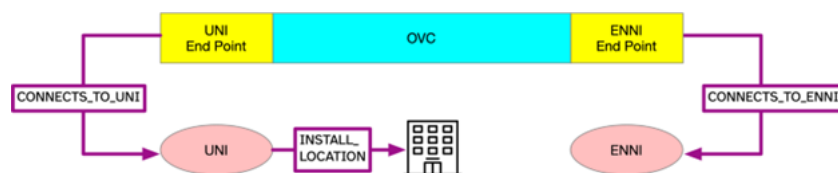


Figure 9. Operator Carrier Ethernet UNI Product Shortened Configuration Example

In case, some of these requirements are violated the Seller returns an error response to the Buyer that indicates specific functional errors. These errors are listed in the response body (a list of **Error422** entries) for HTTP **422** response.

5.6. Model Structural Validation

The structure of the HTTP payloads is defined using:

- OpenAPI version 3.0 for product-agnostic part of the payload
- JsonSchema (draft 7) for product-specific part of the payload

[R7] Implementations **MUST** use payloads that conform to these definitions.

[R8] The Buyer and the Seller **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

[R9] A product specification may define additional consistency rules and requirements that **MUST** be respected by implementations.

These are defined for:

- required relation type, multiplicity to other items in the same quote request
- required relation type, multiplicity to entities in the Seller's product inventory
- related contact information roles that are to be defined at the item level
- relations to places (locations) and their roles that are to be defined at the item level

5.7. Security Considerations

There must be an authentication mechanism whereby a Seller can be assured who a Buyer is and vice-versa. There must also be authorization mechanisms in place to control what a particular Buyer or Seller is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. Security considerations are standardized by *LSO API Security Profile* [[MEF 128.1](#)].

6. API Interaction & Flows

This section provides a detailed insight into the API functionality, use cases, and flows. First, it presents a list of business use cases and then provides examples with a comprehensive explanation of all usage aspects.

Use Case #	Use Case Name	Use Case Description
1	Retrieve Product Offering Availability	The Buyer requests a list of available Product Offering Configurations from the Seller for a specific Product Specification within the specified Delivery Context. The Seller responds to the Buyer with a list of Product Configurations meeting the Buyer's criteria and the Installation Interval for each of these. Each Product Configuration has a unique identifier that is passed to the Buyer by the Seller. This identifier is used to Retrieve Pricing and Terms.
2	Retrieve Pricing and Terms	The Buyer requests a list of Pricing and Terms from the Seller for a specific Product Configuration Identifier within a specific Delivery Context. The Seller responds to the Buyer with a list of Pricing and Terms.

Table 3. Use cases description

The detailed business requirements of each of the use cases are described in section 8 of [MEF110].

6.1. Use case 1: Retrieve Product Offering Availability

To send a Retrieve Product Offering Availability request the Buyer uses the **POST /productOfferingAvailability**. The retrieve operation is performed with a **POST** operation to allow passing complex data structure of the Delivery Context as a request payload.

The flow is a simple request-response pattern, as presented in Figure 10:

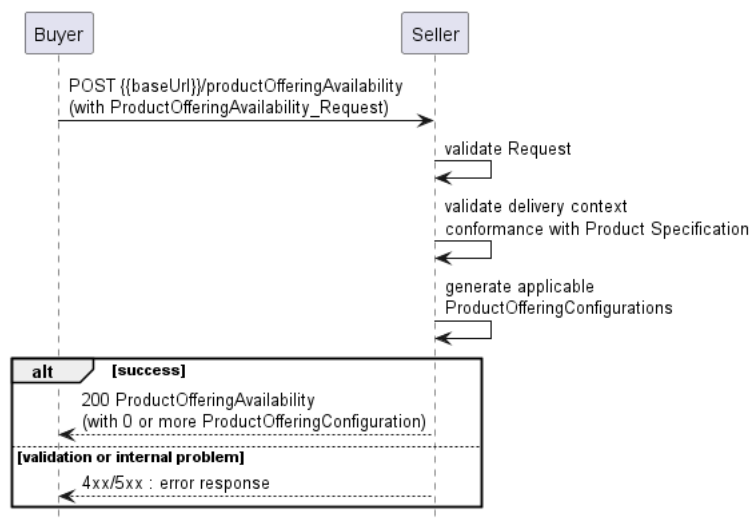


Figure 10. Use Case 1 Flow

This use case is similar to one defined in Product Offering Qualification [MEF 87] but is optimized for more effective discovery of available product configurations. The Buyer provides only the Product Specification and the Delivery Context and receives a list of available product

configurations in the response. In MEF 87 the Buyer provides one product configuration and receives the Seller's response only for this one specific set of attributes (if matched with available configurations). This led to many failed requests until the Buyer got to know the possible configuration available in the given delivery context.

For more details please refer to [MEF 110], section 9.1.

Figure 11 presents the model of Use Case 1. The request uses **ProductOfferingAvailability_Request** as the root object and the response is provided with the use of **ProductOfferingAvailability**.

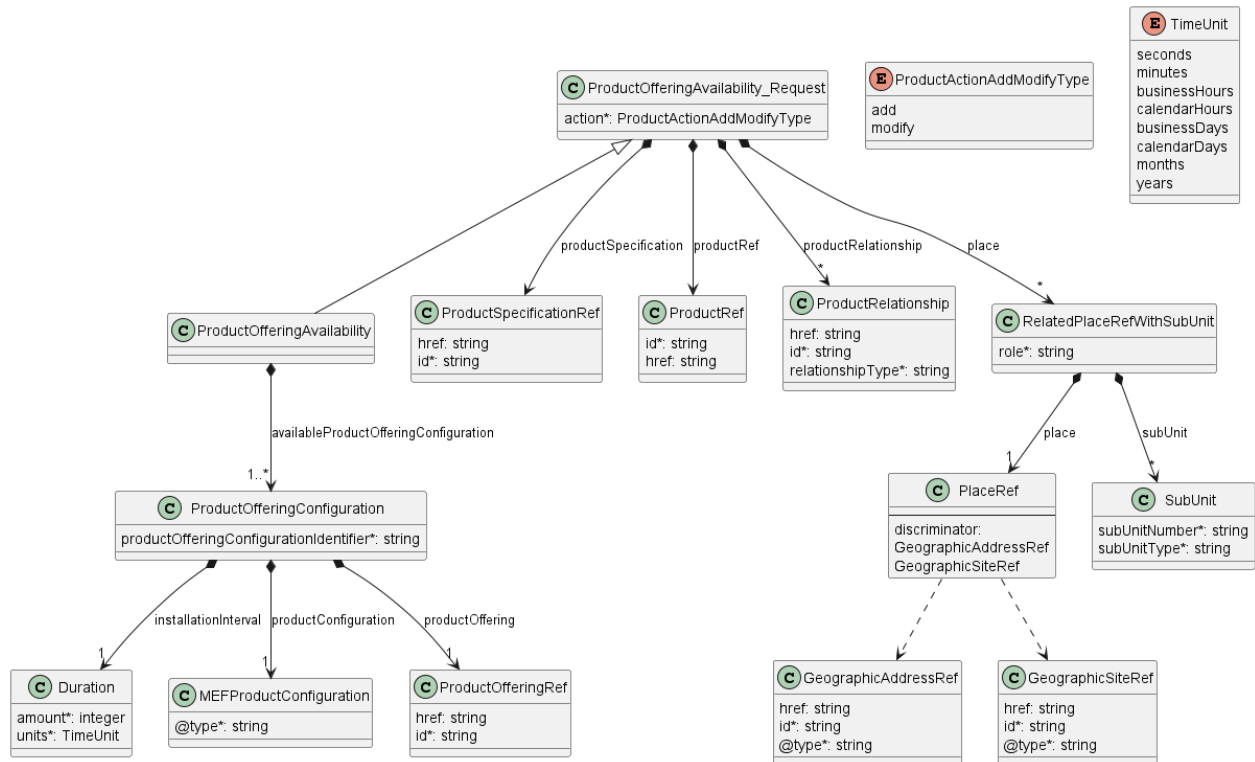


Figure 11. Use Case 1 Product Offering Availability Model

6.1.1. Request

The question that the Buyer asks with this request is "What **availableProductOfferingConfiguration** of product of given **productSpecification** can you provide me with given delivery context?".

The context can be provided by:

- **place** - a reference to a Geographic Address or Site where the product is to be installed. This applies to location-oriented products, e.g. Operator UNI.
- **productRelationship** - relation to an existing product to which the product being asked for references, e.g. Access E-Line OVC pointing to UNI and ENNI products.
- **productRef** - in case of modification request a reference to an existing product that is to be updated. In this case, there is no need to provide additional context.

The following snippet shows the body of a Product Offering Availability check request:

ProductOfferingAvailability_Request:

```
{
  "action": "add",
```

```

"productSpecification": {
  "id": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all"
},
"productRelationship": [
  {
    "id": "UNI-ID-0001",
    "relationshipType": "CONNECTS_TO_UNI"
  },
  {
    "id": "ENNI-ID-0001",
    "relationshipType": "CONNECTS_TO_ENNI"
  }
]
}

```

The request is very simple and has only three parameters:

- **action** - to state if the Buyer is interested to **add** or **modify** a product.
- **productSpecification**: to point to product specification that is of Buyer's interest. For the sake of example, the **id** in the example is the **urn** as defined by MEF standard describing the Access E-Line. In practice, this is an identifier as presented by the Seller in the Product Catalog.
- **productRelationship** - to provide the delivery context. As shown in Figure 9 Access E-Line requires 2 product relationships. Other products may require providing place relationship (e.g. UNI)

An example of a request for a Product that requires providing a relationship to a place (e.g. Operaton UNI) is presented in the following snippet:

ProductOfferingAvailability_Request:

```

{
  "action": "add",
  "productSpecification": {
    "id": "urn:mef:lso:spec:sonata:carrier-ethernet-operator-uni:v5.0.0:all"
  },
  "place": [
    {
      "place": {
        "@type": "GeographicAddressRef",
        "id": "GeographicAddressId-0005"
      },
      "role": "INSTALL_LOCATION"
    }
  ]
}

```

[R10] If **action=add**, the request **MUST** provide **productSpecification**. [MEF110 R3]

[R11] If **action=add**, the request **MUST NOT** provide **productRef**. [MEF110 R7]

[R12] If **action=add** and the Product Specification defines mandatory place relations, the request **MUST** provide them (and only them) accordingly in the **place** attribute. [MEF110 R5], [MEF110 R9]

[R13] When specifying the **place**, the Buyer **MUST** provide following attributes: [MEF110 R48]

- **place**
- **role**

[R14] When specifying the **place**, with **GeographicSiteRef** the Buyer **MUST NOT** additionally provide **subUnit** to describe exactly where the Buyer wants the Product to be installed.

[R15] If **action=add** and the Product Specification defines mandatory product relations, the request **MUST** provide them (and only them) accordingly in the **productRelationship** attribute. [MEF110 R6], [MEF110 R8]

There may be cases, as explicitly allowed or defined in the Product Specification where a Buyer cannot precisely state which installed Product a newly installed Product will be related to because the Buyer will determine this at the time of submitting the Product Order. The Buyer can include a list of candidate installed Products to be validated against. An example is the ENNI for an Access E-Line Product where the Buyer may, for example, include a list of ENNIs between the Buyer and Seller as related Products. The Products in the list would be expected to comply with a business rule agreed to by the Buyer and Seller (e.g. ENNIs that are all live and all in the same Geographic Area as defined by the Seller). In such cases multiple **productRelationships** with the same **role** are provided by the Buyer, as in the following example. When this happens, it is at the Seller's discretion to choose the item on the list which is used to fulfil the request.

```
{
  "action": "add",
  "productSpecification": {
    "id": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all"
  },
  "productRelationship": [
    {
      "id": "UNI-ID-0001",
      "relationshipType": "CONNECTS_TO_UNI"
    },
    {
      "id": "ENNI-ID-0001",
      "relationshipType": "CONNECTS_TO_ENNI"
    },
    {
      "id": "ENNI-ID-0002",
      "relationshipType": "CONNECTS_TO_ENNI"
    },
    {
      "id": "ENNI-ID-0003",
      "relationshipType": "CONNECTS_TO_ENNI"
    }
  ]
}
```

[R16] If **action=modify**, the request **MUST** provide **productRef**. [MEF110 R4]

[R17] If **action=modify**, the request **MUST NOT** provide following attributes: [MEF110 R10]

- **place**
- **productRelationship**
- **productSpecification**

6.1.2. Response

The Seller performs necessary checks for the request's correctness and compliance with provided Product Specifications and provides a response.

An example of a response to an example of a request is presented in the following snippet.

```
{
  "action": "add",
  "productSpecification": {
    "id": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all"
  },
  "productRelationship": [
    {
      "id": "UNI-ID-0001",
```

```

    "relationshipType": "CONNECTS_TO_UNI"
  },
  {
    "id": "ENNI-ID-0001",
    "relationshipType": "CONNECTS_TO_ENNI"
  }
],
"availableProductOfferingConfiguration": [
  {
    "productOffering": {
      "id": "Access E-Line OVC - Low Class of Service"
    },
    "productOfferingConfigurationIdentifier": "POC-ID-0001",
    "productConfiguration": {
      "@type": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all",
      "maximumFrameSize": 1526,
      "listOfClassOfServiceNames": ["low"],
      "enniEp": {
        "identifier": "ENNI-ID-0001-EndPoint-0001"
      },
      "uniEp": {
        "identifier": "UNI-ID-0001-EndPoint-0001",
        "ingressBandwidthProfilePerClassOfServiceName": [
          {
            "classOfServiceName": "low",
            "bwpFlow": {
              "cir": {
                "irValue": 0,
                "irUnits": "MBPS"
              },
              "cirMax": {
                "irValue": 0,
                "irUnits": "MBPS"
              },
              "eir": {
                "irValue": 70,
                "irUnits": "MBPS"
              },
              "eirMax": {
                "irValue": 70,
                "irUnits": "MBPS"
              }
            }
          }
        ]
      }
    },
    "installationInterval": {
      "amount": 0,
      "units": "minutes"
    }
  },
  {
    "productOffering": {
      "id": "Access E-Line OVC - High Class of Service"
    },
    "productOfferingConfigurationIdentifier": "POC-ID-0002",
    "productConfiguration": {
      "@type": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all",
      "maximumFrameSize": 1526,
      "listOfClassOfServiceNames": ["high"],
      "enniEp": {
        "identifier": "ENNI-ID-0001-EndPoint-0001"
      },
      "uniEp": {
        "identifier": "UNI-ID-0001-EndPoint-0001",
        "ingressBandwidthProfilePerClassOfServiceName": [
          {
            "classOfServiceName": "high",
            "bwpFlow": {
              "cir": {
                "irValue": 200,
                "irUnits": "MBPS"
              },
              "cirMax": {
                "irValue": 200,
                "irUnits": "MBPS"
              },
              "eir": {
                "irValue": 0,
                "irUnits": "MBPS"
              }
            }
          }
        ]
      }
    }
  }
]

```



```

    },
    "eirMax": {
      "irValue": 0,
      "irUnits": "MBPS"
    }
  }
}
]
}
},
"installationInterval": {
  "amount": 3,
  "units": "minutes"
}
},
{
  "productOffering": {
    "id": "Access E-Line OVC - High Class of Service"
  },
  "productOfferingConfigurationIdentifier": "POC-ID-0003",
  "productConfiguration": {
    "@type": "urn:mef:lso:spec:sonata:access-eline-ovc:v5.0.0:all",
    "maximumFrameSize": 1526,
    "listOfClassOfServiceNames": ["high"],
    "enniEp": {
      "identifier": "ENNI-ID-0001-EndPoint-0001"
    },
    "uniEp": {
      "identifier": "UNI-ID-0001-EndPoint-0001",
      "ingressBandwidthProfilePerClassOfServiceName": [
        {
          "classOfServiceName": "high",
          "bwpFlow": {
            "cir": {
              "irValue": 1,
              "irUnits": "GBPS"
            },
            "cirMax": {
              "irValue": 1,
              "irUnits": "GBPS"
            },
            "eir": {
              "irValue": 0,
              "irUnits": "MBPS"
            },
            "eirMax": {
              "irValue": 0,
              "irUnits": "MBPS"
            }
          }
        }
      ]
    }
  }
}
],
},
"installationInterval": {
  "amount": 3,
  "units": "minutes"
}
}
]
}

```

Figure 12 presents the general structure of the response. It echoes back the Buyer's request and provides a list of **availableProductOfferingConfiguration**. For better readability the details of **productConfigurations** are "<<skipped>>" and will be presented on further figures.

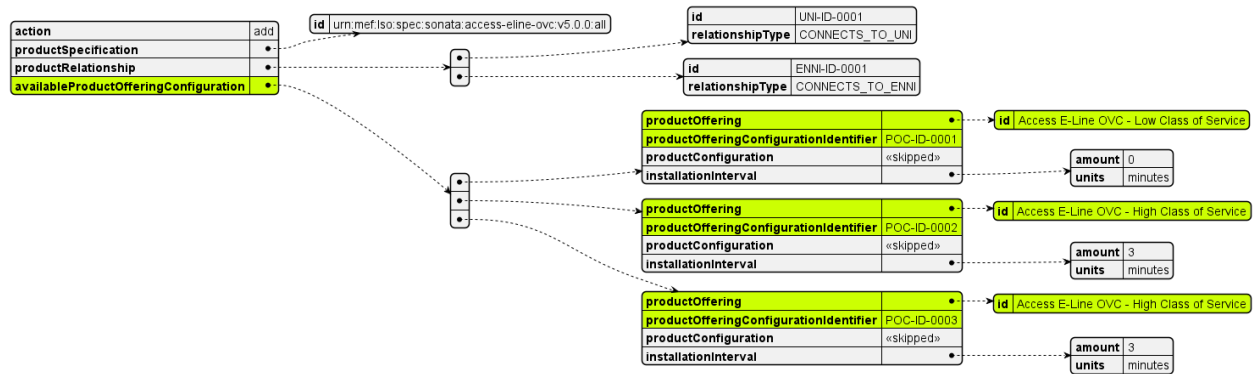


Figure 12. Use Case 1 Response structure

In this example, for given Delivery Context and Product Specification there are 3 **availableProductOfferingConfiguration**: 1 for **Access E-Line OVC - Low Class of Service** Product Offering and with real time delivery (**installationInterval.amount=0**), and 2 for **Access E-Line OVC - High Class of Service** Product Offering with 2 different bandwidth configurations. Each has its own distinct **productOfferingConfigurationIdentifier**.

Note: The **productOfferingConfigurationIdentifier** identifies only the Product Specific Attributes values as provided in the **productConfiguration** attribute. It does not identify the action or the delivery context. This allows to ask for Pricing Discovery of the same **productConfiguration** in different delivery contexts.

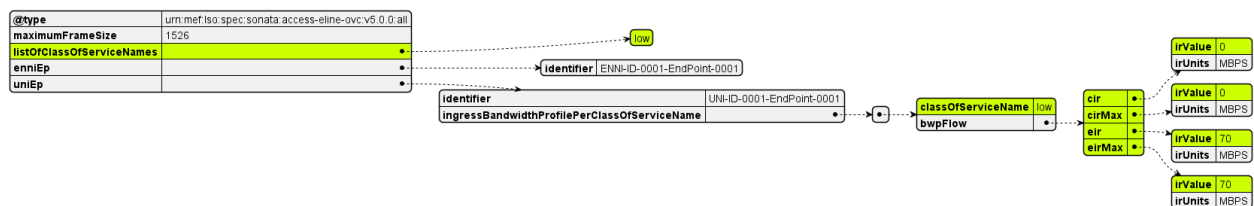


Figure 13. UC1 Response Product Configuration 1

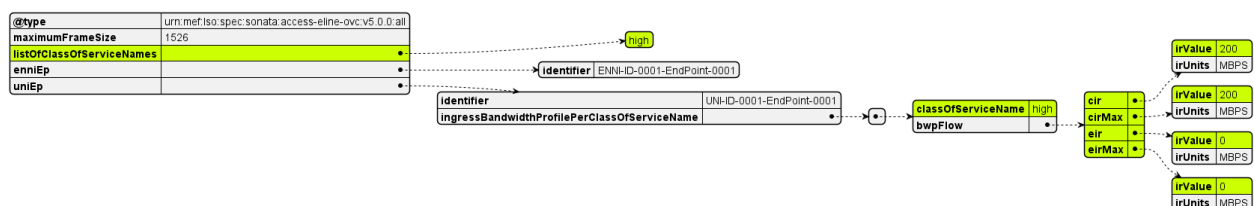


Figure 14. UC1 Response Product Configuration 2

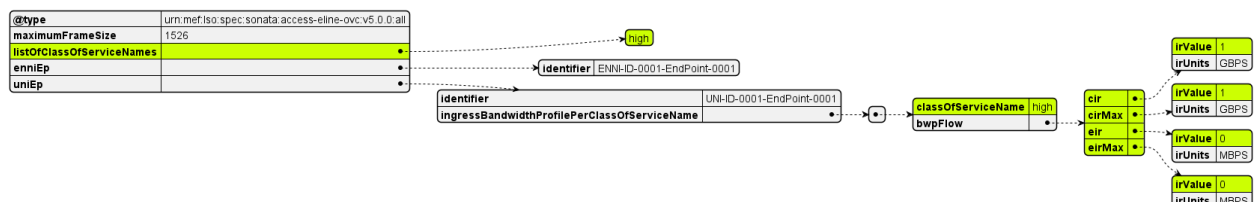


Figure 15. UC1 Response Product Configuration 3

Figures 13, 14, and 15 present details of product configurations and highlight differences between them. The first one has a **low** class of service and has **eir** and **eirMax** attributes set to **70 MBPS**. This means it offers a best-effort bandwidth of a maximum 70 MBPS. The latter have a **high** class of service. The second one has the **cir** and **cirMax** set to **200 MBPS** thus offering a guaranteed bandwidth of **200 MBPS**, while the third one offers **1GBPS** of guaranteed bandwidth.

[R18] The Seller **MUST** echo back the attributes of the Buyer's Product Offering Availability request. [MEF110 R12]

[R19] The Seller **MUST NOT** return a response that contains `productOfferingConfigurations` that differ only by `installationInterval` (in which all other attributes are the same). [MEF110 R13]

In other words - the Seller is expected to provide only the shortest possible `installationInterval` per given `productConfiguration`.

[R20] If the request is successful, the response given by the Seller **MUST** include a list of zero or more `availableProductOfferingConfiguration`. [MEF110 R14]

[R21] If the request is unsuccessful or fails validation, the Seller **MUST** return an `Error` response with the appropriate Error Code. [MEF110 R15], [MEF110 R16]

[R22] For each returned `ProductOfferingConfiguration` the Seller **MUST** include following attributes: [MEF110 R17], [MEF110 R18], [MEF110 R21], [MEF110 R20]

- `installationInterval`
- `productOffering`
- `productConfiguration`
- `productOfferingConfigurationIdentifier`

[R23] Every returned `productOfferingConfigurationIdentifier` **MUST** uniquely identify a `ProductOfferingConfiguration` within the Seller. [MEF110 R17]

[R24] Every returned `productOfferingConfigurationIdentifier` **MUST** be valid for Pricing Discovery request for a period of at least 15 minutes. [MEF110 R23]

[R25] Every returned `productConfiguration` **MUST** contain only attributes specified by Product Specification and agreed to by the Buyer and Seller during the onboarding. [MEF110 R19], [MEF110 R20]

6.2. Use case 2: Retrieve Pricing for a Product Offering Configuration

This Use Case allows the Buyer to perform a Pricing Discovery (ask for Pricing and Terms) for a specific Product Configuration identified by `productOfferingConfigurationIdentifier` (most probably) obtained in the previous Use Case.

This can be done with the use of the `POST /pricingDiscovery` operation. The retrieve operation is performed with a `POST` operation to allow passing the complex data structure of the Delivery Context as a request payload.

The flow is a simple request-response pattern, as presented in Figure 16:

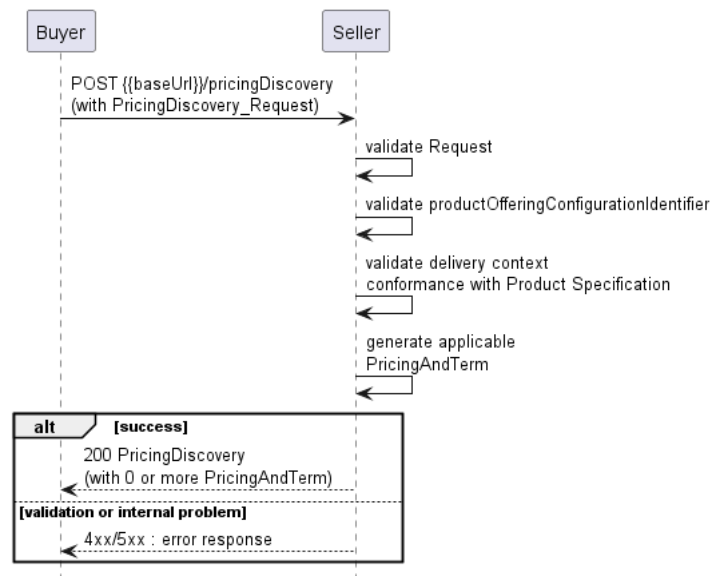


Figure 16. Pricing Discovery Flow

This use case is similar to one defined in Quote API [MEF 115] but is optimized for more effective discovery of available pricing and terms. The Buyer provides only the `productOfferingConfigurationIdentifier` and the Delivery Context and receives a list of available pricings and terms in the response. In MEF 115 the Buyer provides one full product configuration (by value) and term and receives the Seller's response only for this one specific combination (if matched with available configurations). This requires a request per each term, assuming the Buyer knows exactly what are the available terms, or a set of tries until the Buyer discovers all available terms.

Figure 17 presents the model of Use Case 1. The request uses `PricingDiscovery_Request` as the root object and the response is provided with use of `PricingDiscovery`.

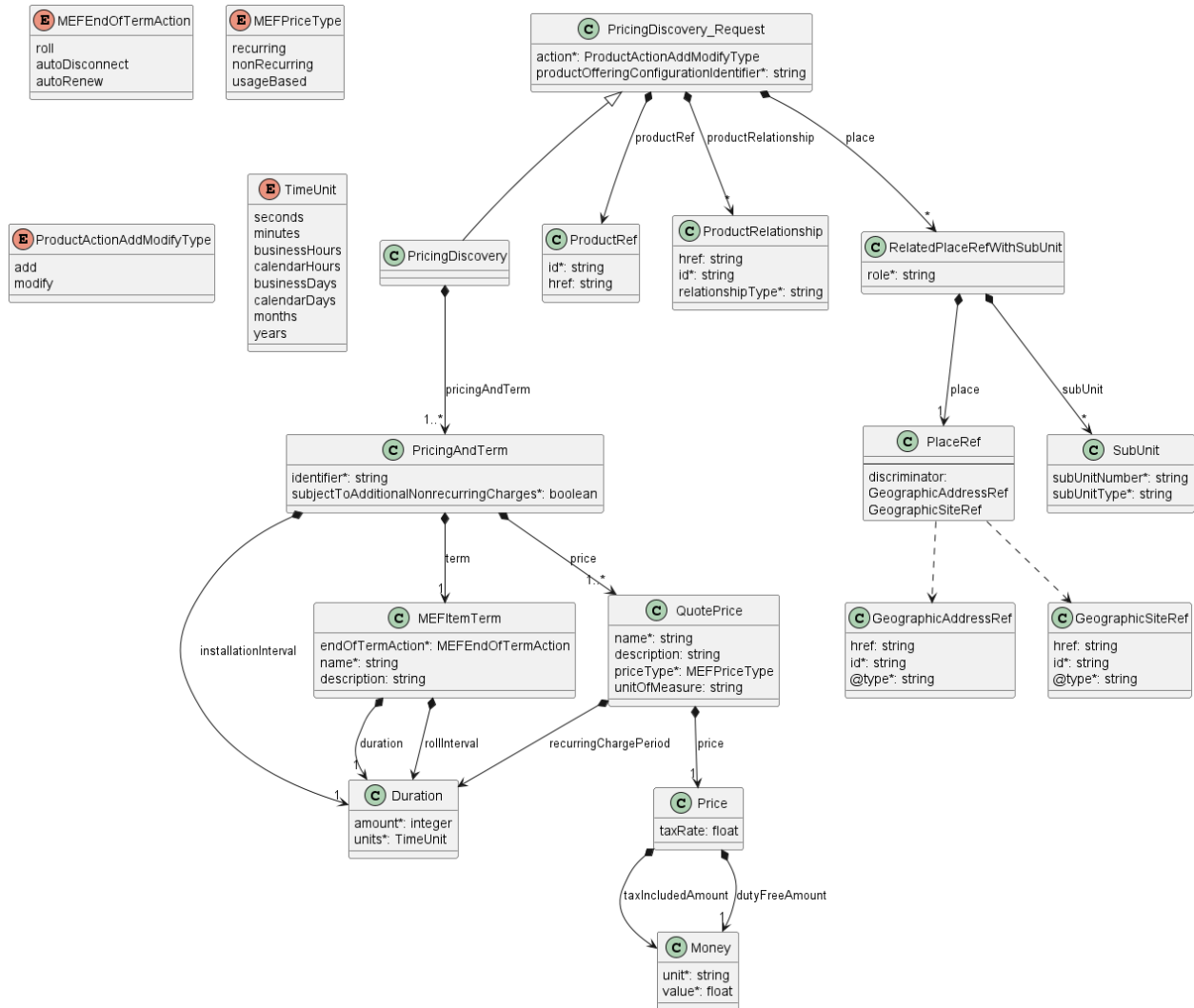


Figure 17. Use Case 2 Pricing Discovery Model

6.2.1. Request

The question that the Buyer asks with this request is "What **PricingAndTerms** for a **productOfferingConfigurationIdentifier** can you provide me with given delivery context?".

The rules of providing the delivery context are the same as described in Use Case 1.

The following snippet shows the body of a Price Discovery request:

PriceDiscovery_Request:

```
{
  "action": "add",
  "productRelationship": [
    {
      "id": "UNI-ID-0001",
      "relationshipType": "CONNECTS_TO_UNI"
    },
    {
      "id": "ENNI-ID-0001",
      "relationshipType": "CONNECTS_TO_ENNI"
    }
  ],
  "productOfferingConfigurationIdentifier": "POC-ID-0002"
}
```

The request is very simple and has only three parameters:

- **action** - to state if the Buyer wishes to **add** or **modify** a product.
- **productOfferingConfigurationIdentifier**- to refer to Product Offering Configuration that the Buyer is willing to price
- **productRelationship** - to provide the delivery context. As shown on Figure 9 Access E-Line requires 2 product relationships. Other products may require providing place relationship (e.g. UNI)

[R26] The request **MUST** provide: [MEF110 R26], [MEF110 R28]

- **productOfferingConfigurationIdentifier**
- **action**

[R27] If **action=add** and the Product Specification (derived from **productOfferingConfigurationIdentifier**) defines mandatory place relations, the request **MUST** provide them (and only them) accordingly in the **place** attribute. [MEF110 R26], [MEF110 R30]

[R28] If **action=add** and the Product Specification (derived from **productOfferingConfigurationIdentifier**) defines mandatory product relations, the request **MUST** provide them (and only them) accordingly in the **productRelationship** attribute. [MEF110 R26], [MEF110 R31]

[R29] If **action=add**, the request **MUST NOT** provide **productRef**. [MEF110 R27]

[R30] If **action=modify**, the request **MUST** additionally provide **productRef**. [MEF110 R28]

[R31] If **action=modify**, the request **MUST NOT** provide: [MEF110 R29]

- **place**
- **productRelationship**

6.2.2. Response

The Seller performs necessary checks for the request's correctness, compliance with provided Product Specification, and **productOfferingConfigurationIdentifier** validity and then provides a response.

An example of a response to an example of a request is presented in the following snippet:

```
{
  "action": "add",
  "productRelationship": [
    {
      "id": "UNI-ID-0001",
      "relationshipType": "CONNECTS_TO_UNI"
    },
    {
      "id": "ENNI-ID-0001",
      "relationshipType": "CONNECTS_TO_ENNI"
    }
  ],
  "productOfferingConfigurationIdentifier": "POC-ID-0002",
  "pricingAndTerm": [
    {
      "term": {
        "duration": {
          "amount": 12,
          "units": "months"
        },
        "endOfTermAction": "roll",
      }
    }
  ]
}
```

```

    "name": "1-year term",
    "rollInterval": {
      "amount": 1,
      "units": "months"
    }
  },
  "validFor": {
    "startDateTime": "2023-06-02T12:24:48.687Z",
    "endDateTime": "2023-06-09T12:24:48.687Z"
  },
  "subjectToAdditionalNonrecurringCharges": false,
  "price": [
    {
      "price": {
        "taxRate": 10,
        "taxIncludedAmount": {
          "unit": "EUR",
          "value": 110
        },
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 100
        }
      },
      "name": "Monthly price for a 1-year subscription",
      "priceType": "recurring",
      "recurringChargePeriod": {
        "amount": 1,
        "units": "calendarMonths"
      }
    }
  ],
  "firm": true,
  "installationInterval": {
    "amount": 3,
    "units": "minutes"
  }
},
{
  "term": {
    "duration": {
      "amount": 36,
      "units": "months"
    },
    "endOfTermAction": "roll",
    "name": "3-year term",
    "rollInterval": {
      "amount": 1,
      "units": "months"
    }
  },
  "validFor": {
    "startDateTime": "2023-06-02T12:24:48.687Z",
    "endDateTime": "2023-06-09T12:24:48.687Z"
  },
  "subjectToAdditionalNonrecurringCharges": false,
  "price": [
    {
      "price": {
        "taxRate": 10,
        "taxIncludedAmount": {
          "unit": "EUR",
          "value": 88
        },
        "dutyFreeAmount": {
          "unit": "EUR",
          "value": 80
        }
      },
      "name": "Monthly price for a 3-year subscription",
      "priceType": "recurring",
      "recurringChargePeriod": {
        "amount": 1,
        "units": "calendarMonths"
      }
    }
  ],
  "firm": true,
  "installationInterval": {
    "amount": 3,
    "units": "minutes"
  }
}

```

```

    }
  }
}

```

Figure 18 presents the Seller's response. It echoes back the Buyer's request and provides a list of **pricingAndTerm**. In this example in the given Delivery Context and **productOfferingConfigurationIdentifier** the Seller has two possible **PricingAndTerm**. Differences between them are highlighted. The first one is for a 1-year contract with a 110 EUR monthly rate, the other is for a 3-year contract with a 88 EUR monthly rate. Each has its own distinct **identifier**.

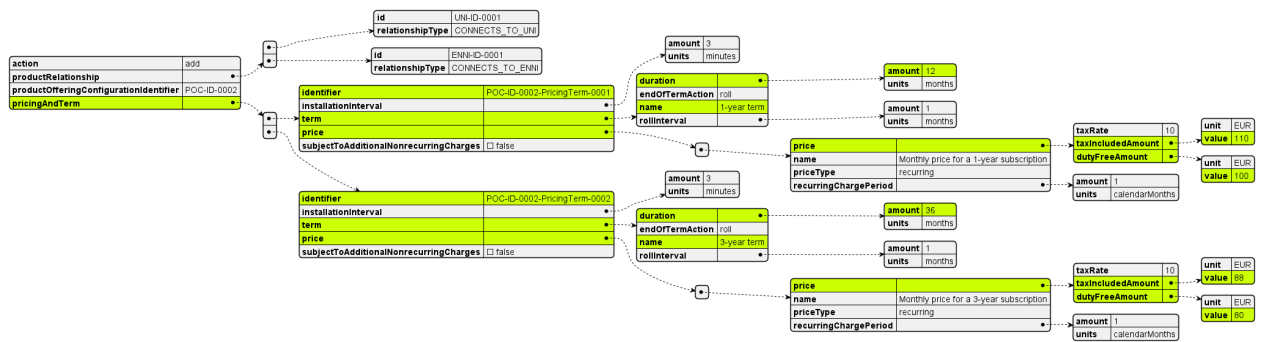


Figure 18. Use Case 2 Response

[R32] If the request is unsuccessful or fails validation, the Seller **MUST** return an **Error** response with the appropriate Error Code. [MEF110 R35]

[R33] If the **productOfferingConfigurationIdentifier** has expired, the Seller **MUST** return an error. [MEF110 R33]

[R34] The Seller **MUST** echo back the attributes of the Buyer's Pricing Discovery request. [MEF110 R34]

[R35] If the request is successful, the response given by the Seller **MUST** include a list of zero or more **pricingAndTerm** items. [MEF110 R36]

[R36] For each returned **PricingAndTerm** the Seller **MUST** include following attributes: [MEF110 R39]

- **identifier**
- **term**
- **price**
- **subjectToAdditionalNonrecurringCharges**
- **installationInterval**

[R37] When providing **MEFItemTerm**, following attributes **MUST** be included: [MEF110 R39]

- **duration**
- **endOfTermAction**

[R38] Once assigned **PricingAndTerm.identifier** **MUST** be be unique within the Seller's system. [MEF110 R45]

[R39] **PricingAndTerm** provided by the Seller **MUST** be valid for at least 15 minutes. [MEF110 R42]

[R409] If the `installationInterval` is 0 (immediate) the `subjectToAdditionalNonrecurringCharges` **MUST** be set to `false`. [MEF110 R43]

The `subjectToAdditionalNonrecurringCharges` attribute is used to indicate additional construction costs or other one-time costs that are not known may be added during fulfillment. This is managed through the Charge Process in the Product Order Process.

If the `subjectToAdditionalNonrecurringCharges` attribute value is `false`, then this is equivalent to the Firm value in MEF 115, which means that all specified monthly recurring Charges and non-recurring Charges are committed. If the `subjectToAdditionalNonrecurringCharges` attribute value of the flag is `true`, then this is equivalent to the Subject to Feasibility Check value in MEF 115 and states that provided monthly recurring charges are final but non-recurring charges are subject to change during fulfillment.

[R41] If the `term.endOfTermAction` is set to `roll` the Seller **MUST** provide the `term.rollInterval` attribute. [MEF110 R40]

[R42] If the `term.endOfTermAction` is set to `autoDisconnect` or `autoRenew` the Seller **MUST NOT** provide the `term.rollInterval` attribute. [MEF110 R41]

[R43] If a Seller has returned multiple `productOfferingConfigurationIdentifiers` to the Buyer in different Product Offering Availability responses that refer to the same `productConfiguration` values, then the Seller **MUST** return the same response to a Pricing Discovery request for a given delivery context that contains any of those `productOfferingConfigurationIdentifiers` that are within their validity period. [MEF110 R38]

[R43] means that a Retrieve Pricing and Terms request for a given Delivery Context and a given `productConfiguration` values always returns the same result regardless of which `productOfferingConfigurationIdentifier` is used to refer to that `ProductOfferingConfiguration`. In particular, this is the case even if the `productOfferingConfigurationIdentifier` was originally returned to the Buyer in an Availability Discovery response for a different Delivery Context.

The following requirements apply to every item returned in the `price` list of the response.

[R44] The Seller **MUST** provide at least one `price` item of `priceType=recurring` if a recurring charge applies. [MEF80 R55]

[R45] For each provided `price` item (`QuotePrice`) the Seller **MUST** include the following attributes: [MEF80 R55]

- `name`
- `priceType`
- `price`

[R46] The `recurringChargePeriod` **MUST** only be provided if the `priceType` is `recurring`. [MEF80 R56]

[R47] The `unitOfMeasure` **MUST** only be provided if the `priceType` is `usageBased`. [MEF80 R57]

Table 4 shows the combination of attributes that must be provided for each `priceType`:

<code>priceType</code>	<code>recurringChargePeriod</code>	<code>unitOfMeasure</code>	<code>price.dutyFreeAmount</code>	Comments
<code>recurring</code>	X		X	
<code>nonRecurring</code>			X	

priceType	recurringChargePeriod	unitOfMeasure	price.dutyFreeAmount	Comments
usageBased		X	X	price.dutyFreeAmount is the charge for unitOfMeasure

Table 4. Price Type Required Information

7. API Details

7.1. API patterns

7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Product Order API uses the error responses as depicted and described below.

Implementations can use HTTP error codes not specified in this standard in compliance with rules defined in RFC 7231 [RFC7231]. In such a case, the error message body structure might be aligned with the **Error**.

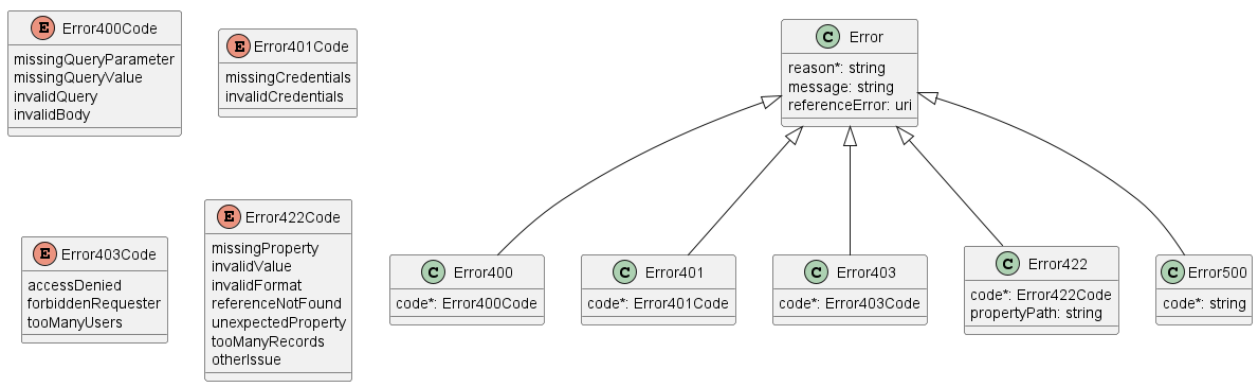


Figure 19. Data model types to represent an erroneous response

7.1.1.1. Type Error

Description: Standard Class used to describe API response error Not intended to be used directly. The **code** in the HTTP header is used as a discriminator for the type of error returned in runtime.

Name	Type	Description
reason*	string maxLength = 255	Text that explains the reason for the error. This can be shown to a client user.
message	string	Text that provides mode details and corrective actions related to the error. This can be shown to a client user.
referenceError	uri format = uri	URL pointing to documentation describing the error

7.1.1.2. Type Error400

Description: Bad Request. (<https://tools.ietf.org/html/rfc7231#section-6.5.1>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

Name	Type	Description
code*	Error400Code	One of the following error codes: - missingQueryParameter: The URI is missing a required query-string parameter - missingQueryValue: The URI is missing a required query-string parameter value - invalidQuery: The query section of the URI is invalid. - invalidBody: The request has an invalid body

7.1.1.3. **enum** Error400Code

Description: One of the following error codes:

- missingQueryParameter: The URI is missing a required query-string parameter
- missingQueryValue: The URI is missing a required query-string parameter value
- invalidQuery: The query section of the URI is invalid.
- invalidBody: The request has an invalid body

7.1.1.4. **Type** Error401

Description: Unauthorized. (<https://tools.ietf.org/html/rfc7235#section-3.1>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error401Code	One of the following error codes: - missingCredentials: No credentials provided. - invalidCredentials: Provided credentials are invalid or expired

7.1.1.5. **enum** Error401Code

Description: One of the following error codes:

- missingCredentials: No credentials provided.
- invalidCredentials: Provided credentials are invalid or expired

7.1.1.6. **Type** Error403

Description: Forbidden. This code indicates that the server understood the request but refuses to authorize it. (<https://tools.ietf.org/html/rfc7231#section-6.5.3>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

Name	Type	Description
code*	Error403Code	This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes: - accessDenied: Access denied - forbiddenRequester: Forbidden requester - tooManyUsers: Too many users

7.1.1.7. **enum** Error403Code

Description: This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- accessDenied: Access denied
- forbiddenRequester: Forbidden requester
- tooManyUsers: Too many users

7.1.1.8. **Type** Error422

Description: Unprocessable entity due to a business validation problem. (<https://tools.ietf.org/html/rfc4918#section-11.2>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error422Code	One of the following error codes: - missingProperty: The property the Seller has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the Seller system - unexpectedProperty: Additional property, not expected by the Seller has been provided - tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason)
propertyPath	string	A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer (https://tools.ietf.org/html/rfc6901).

7.1.1.9. **enum** Error422Code

Description: One of the following error codes:

- missingProperty: The property the Seller has expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format

- **referenceNotFound:** The object referenced by the property cannot be identified in the Seller system
- **unexpectedProperty:** Additional property, not expected by the Seller has been provided
- **tooManyRecords:** the number of records to be provided in the response exceeds the Seller's threshold.
- **otherIssue:** Other problem was identified (detailed information provided in a reason)

7.1.1.10. Type Error500

Description: Internal Server Error. (<https://tools.ietf.org/html/rfc7231#section-6.6.1>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	string	- internalError: Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request.

The following error code:

7.2. API Data model

7.2.1. Product Offering Availability

7.2.1.1. Type ProductOfferingAvailability_Request

Description: Product Offering Availability Request allows the Buyer to provide the action, delivery context, and Product Specification to ask the Seller to provide a list of available Product Offering Configurations.

Name	Type	M/O	Description	MEF 110
action	ProductActionAddModifyType	M	The action to be performed by the Seller to fulfill any Product Order that results from this interaction.	Action
productSpecification	ProductSpecificationRef	O	A reference to a Product Specification used to describe the Product. This MUST be provided when the `action` is `add`. It MUST NOT be provided when the `action` = `modify`	Product Specification Identifier

Name	Type	M/O	Description	MEF 110
productRef	ProductRef	O	A reference to a Product in the Product Inventory. It MUST NOT be provided when the `action` = `add`. It MUST be provided when the `action` = `modify`.	Product Identifier
productRelationship	ProductRelationship[]	O	A list of Product Relationships as defined by the Product Specification. It MUST be provided when the `action` = `add`. It MUST NOT be provided when the `action` = `modify`.	Product Relationships
place	RelatedPlaceRefWithSubUnit[]	O	A list of Geographic Addresses or Sites and their roles in relation to the Product. It MUST be provided when the `action` = `add`. It MUST NOT be provided when the `action` = `modify`.	Place Relationships

7.2.1.2. Type ProductOfferingAvailability

Description: Product Offering Availability allows the Seller to provide a list of Product Offering Configurations matching action, delivery context and Product Specification provided by the Buyer

Inherits from:

- [ProductOfferingAvailability_Request](#)

Name	Type	M/O	Description	MEF
------	------	-----	-------------	-----

Name	Type	M/O	Description	MEF
availableProductOfferingConfiguration	ProductOfferingConfiguration [] minItems = 0	M	The list of available Product Offering Configurations	Avai Prod Offe Con

7.2.1.3. Type ProductOfferingConfiguration

Description: Allows the Seller to provide detailed information of the Product Configuration matching Buyer's Product Offering Availability Request.

Name	Type	M/O	Description	M
productOffering	ProductOfferingRef	M	The identifier of the Product Offering for which this Product Configuration is valid.	P C I
productConfiguration	MEFProductConfiguration	M	The set of technical attributes for the Product Offering that make this configuration unique. This essentially specifies the values for attributes defined in the Product Offering.	P S A
productOfferingConfigurationIdentifier	string	M	An identifier of the returned 'productConfiguration' that can be used in the next step for Pricing Discovery	P C C I
installationInterval	Duration	M	The indicative duration (>=0) for the delivery of the configuration in the specified Delivery Context. The shortest possible interval is specified by the Seller. It is not considered a commitment by the Seller.	I I

7.2.1.4. Type ProductOfferingRef

Description: A reference to a Product Offering offered by the Seller to the Buyer.

Name	Type	M/O	Description	MEF 110
------	------	-----	-------------	---------

Name	Type	M/O	Description	MEF 110
href	string	O	Hyperlink to a Product Offering in the Seller catalog. In case the Seller is not providing a catalog API this field is not used. The catalog is provided by the Seller to the Buyer during onboarding. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	Not represented in MEF 110
id	string	M	id of a Product Offering. It is assigned by the Seller. The Buyer and the Seller exchange information about offerings' ids during the onboarding process.	Product Offering Identifier

7.2.1.5. Type MEFProductConfiguration

Description: MEFProductConfiguration is used as an extension point for MEF specific product/service payload. The **@type** attribute is used as a discriminator

Name	Type	M/O	Description	MEF 110
@type	string	M	The name of the type that uniquely identifies the type of the product that is the subject of the POQ Request. In the case of MEF product this is the URN provided in the Product Specification.	Not represented in MEF 110

7.2.1.6. Type ProductSpecificationRef

Description: A reference to a structured set of well-defined technical attributes and/or behaviors that are used to construct a Product Offering for sale to a market.

Name	Type	M/O	Description	MEF 110
href	string	O	Hyperlink to a Product Specification in the seller's catalog. In case Seller is not providing a catalog API this field is not used. The catalog is provided by the Seller to the Buyer during onboarding. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	Not represented in MEF 110
id	string	M	Unique identifier of the Product Specification	Product Specification Identifier

7.2.2. Pricing Discovery

7.2.2.1. Type PricingDiscovery_Request

Description: A set of attributes common to PricingDiscovery_Request and PricingDiscovery

Name	Type	M/O	Description	MEF
------	------	-----	-------------	-----

Name	Type	M/O	Description	MEF
action	ProductActionAddModifyType	M	The action to be performed by the Seller to fulfill any Order that results from this interaction.	Action
productOfferingConfigurationIdentifier	string	M	The identifier for the Product Offering Configuration that the Pricing Discovery is being requested for.	Product Offering Configuration Identifier
productRef	ProductRef	O	A reference to a Product in the Product Inventory. It MUST NOT be provided when the `action` = `add`. It MUST be provided when the `action` = `modify`.	Product Identifier
productRelationship	ProductRelationship[]	O	A list of Product Relationships as defined by the Product Specification. It MUST be provided when the `action` = `add`. It MUST NOT be provided when the `action` = `modify`.	Product Relationship

Name	Type	M/O	Description	MEF
place	RelatedPlaceRefWithSubUnit[]	O	A list of Geographic Addresses or Sites and their roles with relation to the Product. It MUST be provided when the 'action' = 'add'. It MUST NOT be provided when the 'action' = 'modify'	Place Relati

7.2.2.2. Type PricingDiscovery

Description: PricingDiscovery allows the Seller to provide the list of Pricing and Terms matching action, delivery context, and Product Configuration Identifier provided by the Buyer

Inherits from:

- [PricingDiscovery_Request](#)

Name	Type	M/O	Description	MEF 110
pricingAndTerm	PricingAndTerm[]	M	A set of contract terms and corresponding prices available for a given Product Configuration considering the Delivery Context.	Product Offering Configuration Pricing and Terms

7.2.2.3. Type PricingAndTerm

Description: An option for price and term available to a Buyer for a Product Configuration with a specific Delivery Context

Name	Type	M/O	Description
identifier	string	M	The identifier of the Pricing and Ter

Name	Type	M/O	Description
installationInterval	Duration	M	The indicative duration ≥ 0 for delivery of the configuration in specified Delivery Context when no work is required. The shortest interval specified by the Seller. It is considered a commitment by the Seller. A value of 0 indicates immediate installation.
term	MEFItemTerm	M	The minimum length of the commitment for this PricingAndTerm for the given 'productOfferingConfigurationIdentifier' within the requested Delivery Context.
price	QuotePrice[] <i>minItems = 1</i>	M	The prices that apply to PricingAndTerm for the given Product Offering Configuration within requested Delivery Context.
subjectToAdditionalNonrecurringCharges	boolean	M	An indicator to inform the Buyer additional non-recurring charges may be added during fulfillment.

7.2.2.4. Type QuotePrice

Description: Description of price and discount awarded

Name	Type	M/O	Description	MEF 110
unitOfMeasure	string	O	Unit of Measure if price depending on it (Gb, SMS volume, etc..) MEF: if Quote Item Price Type equals usageBased	Quote Item Price Unit of Measure
price	Price	M	The associated price	Quote Item Price Amount
name	string	M	Name of the quote/quote item price	Quote Item Price Name
priceType	MEFPriceType	M	Indicates if the price is for recurring, non-recurring, or usage based charges	Quote Item Price Type
description	string	O	Description of the quote/quote item price	Quote Item Price Description
recurringChargePeriod	Duration	O	Used for a recurring charge to indicate a period	Quote Item Price Recurring Charge Period

7.2.2.6. Type MEFItemTerm

Description: Describes a term

Name	Type	M/O	Description	MEF 110
name	string	M	Name of the term	Not represented in MEF 110
description	string	O	Description of the term	Description
duration	Duration	M	Duration of the term	Duration
endOfTermAction	MEFEndOfTermAction	M	The action the Seller will take once the term expires.	End of Term Action
rollInterval	Duration	O	The period that the Buyer is required to commit to pay in a recurring fashion at the end of the term to extend the term. If `endOfTermAction` is equal to `roll` then `rollInterval` MUST be specified. If `endOfTermAction` is equal to `autoRenew` or `autoDisconnect`, then `rollInterval` MUST NOT be specified.	Roll Interval

7.2.2.7. enum MEFEndOfTermAction

Description: The action that needs to be taken by the Seller once the term expires.

Value	Description
roll	The Product's contract will continue on a rolling basis once the contract's current term expires
autoDisconnect	The Product will automatically be disconnected (and contract terminated) by the Seller once the contract term expires
autoRenew	The Product's contract will be renewed for another term equivalent to the original contract term.
Value	MEF 110
roll	ROLL
autoDisconnect	AUTO_DISCONNECT
autoRenew	AUTO_RENEW

7.2.2.8. enum MEFPriceType

Description: Indicates if the price is for recurring or non-recurring charges.

Value	MEF 110
recurring	RECURRING
nonRecurring	NON_RECURRING
usageBased	USAGE_BASED

7.2.2.9. Type Money

Description: A base/value business entity used to represent money

Name	Type	M/O	Description	MEF 110
unit	string	M	Currency (ISO4217 norm uses 3 letters to define the currency)	Currency
value	float format = float	M	A positive floating point number	Value

7.2.2.10. Type Price

Description: Provides all amounts (tax included, duty-free, tax rate), used currency and percentage to apply for Price Alteration.

Name	Type	M/O	Description	MEF 110
taxRate	float format = float	O	Price Tax Rate. Unit: [%]. E.g. value 16 stand for 16% tax.	Price Tax Rate
taxIncludedAmount	Money	O	All taxes included amount (expressed in the given currency)	Price Tax Included Amount
dutyFreeAmount	Money	M	All taxes excluded amount (expressed in the given currency)	Price Duty Free Amount

7.2.3. Place representation

7.2.3.1. Type RelatedPlaceRefWithSubUnit

Description: Allows pointing to a place by referring to a GeographicAddress, GeographicSite.

Name	Type	M/O	Description	MEF 110
place	PlaceRef	M		
role	string	M	Role of this place. The values that can be specified here are described by Product Specification (e.g. "INSTALL_LOCATION").	
subUnit	SubUnit[]	O	A list of zero or more sub units included within the boundary of the `place` for this POQ Item. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A. It MUST only be used together with GeographicAddressRef	

7.2.3.2. Type PlaceRef

Description: A place described by reference to Geographic Address or Geographic Site.

7.2.3.3. Type GeographicAddressRef

Description: A reference to a Geographic Address resource available through Address Validation API.

Name	Type	M/O	Description	MEF 110
href	string	O	Hyperlink to the referenced Address. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	
id	string	M	Identifier of the referenced Geographic Address. This identifier is assigned during a successful address validation request (Geographic Address Management API)	
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.3.4. Type GeographicSiteRef

Description: A reference to a Geographic Site resource available through Service Site API

Name	Type	M/O	Description	MEF 110
href	string	O	Hyperlink to the referenced Site. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	Not represented in MEF 110
id	string	M	Identifier of the referenced Geographic Site.	Place Identifier
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.3.5. Type SubUnit

Description: Allows for sub unit identification

Name	Type	M/O	Description	MEF 110
subUnitNumber	string	M	The discriminator used for the subunit, often just a simple number but may also be a range.	
subUnitType	string	M	The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.	

7.2.4. Common Types

7.2.4.1. Type Duration

Description: A Duration in a given unit of time e.g. 3 hours, or 5 days.

Name	Type	M/O	Description	MEF 110
amount	integer minimum = 0	M	Duration (number of seconds, minutes, hours, etc.)	Amount
units	TimeUnit	M	Time unit enumerated	Units

7.2.4.2. enum ProductActionAddModifyType

Description: Action to be performed on the Product.

The following mapping has been used between ProductActionType and MEF 110:

ProductActionType	MEF 110
add	INSTALL
modify	CHANGE
Value	MEF 110
add	ADD
modify	MODIFY

7.2.4.3. Type ProductRef

Description: A reference to a Product in the Product Inventory

Name	Type	M/O	Description	MEF 110
id	string	M	Unique identifier of a Product	Product Identifier
href	string	O	Hyperlink to a Product instance in Sellers Product Inventory. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	Not represented in MEF 110

7.2.4.4. Type ProductRelationship

Description: A relationship to an existing Product. The requirements for usage for given Product are described in the Product Specification.

Name	Type	M/O	Description	MEF 110
href	string	O	Hyperlink to the product in Seller's inventory that is referenced Hyperlink MAY be used when providing a response by the Seller Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request	Not represented in MEF 110

Name	Type	M/O	Description	MEF 110
id	string	M	Unique identifier of the related Product	Related Product Identifier
relationshipType	string	M	Specifies the type (nature) of the relationship to the related Product. The nature of required relationships varies for Products of different types. For example, a UNI or ENNI Product may not have any relationships, but an Access E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Products such as multipoint IP or Firewall Products may have more complex relationships. As a result, the allowed and mandatory `relationshipType` values are defined in the Product Specification.	Relationship Nature

7.2.4.5. **enum** TimeUnit

Description: Represents a unit of time.

Value	MEF 110
seconds	SECONDS
minutes	MINUTES
businessHours	BUSINESS_HOURS
calendarHours	CALENDAR_HOURS
businessDays	BUSINESS_DAYS
calendarDays	CALENDAR_DAYS
months	MONTHS
years	YEARS

8. References

- [MEF 55.1](#) Lifecycle Service Orchestration (LSO): Reference Architecture and Framework, February 2021
- [MEF 55.1.1](#), Amendment to MEF 55.1: Reference Architecture and Framework - Terminology, June 2023
- [MEF 80](#), Quote Management Requirements and Use Cases, July 2021
- [MEF 87](#), LSO Cantata and LSO Sonata Product Offering Qualification API - Developer Guide, May 2022
- [MEF 106](#), LSO Sonata Access E-Line Product Schemas and Developer Guide, February 2023
- [MEF 110](#) Product Offering Availability and Pricing Discovery - Business Requirements and Use Cases, Draft Standard 4, June 2024
- [MEF 115](#) LSO Cantata and LSO Sonata Quote Management API - Developer Guide, May 2022
- [MEF 128.1](#), LSO API Security Profile, April 2024
- [MEF 150](#), Installation Place and Service Site Management Business Requirements and Use Cases, November 2024, Draft Standard (R1)
- [RFC 2119](#), Key words for use in RFCs to Indicate Requirement Levels, March 1997
- [RFC 3986](#) Uniform Resource Identifier (URI): Generic Syntax, January 2005
- [RFC 7231](#), Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content, June 2014 <https://tools.ietf.org/html/rfc7231>
- [RFC 8174](#), Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words, May 2017